

# Logarithmic functions

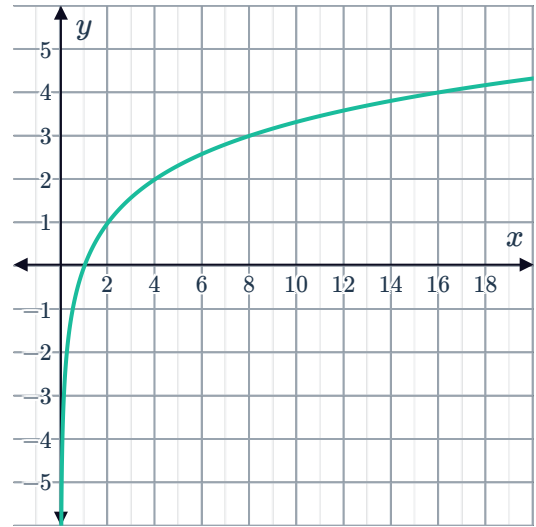


1

a

|     |               |   |   |   |    |
|-----|---------------|---|---|---|----|
| $x$ | $\frac{1}{2}$ | 1 | 2 | 4 | 16 |
| $y$ | -1            | 0 | 1 | 2 | 4  |

b

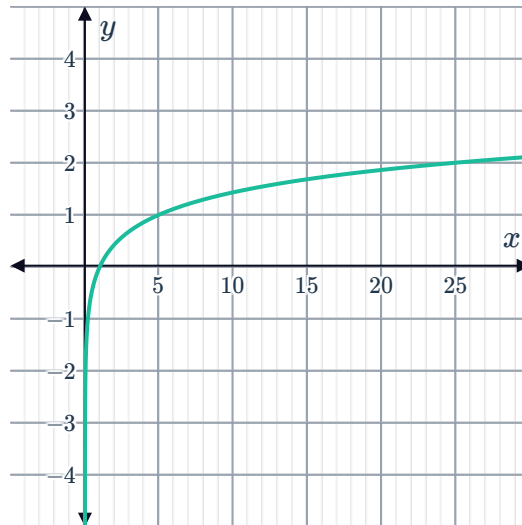


c  $x = 0$

d

- i No    ii No    iii Yes    iv No  
v No

2



3



a

|     |                 |               |   |   |   |    |
|-----|-----------------|---------------|---|---|---|----|
| $x$ | $\frac{1}{243}$ | $\frac{1}{3}$ | 1 | 3 | 9 | 81 |
| $y$ | -5              | -1            | 0 | 1 | 2 | 4  |

b Increasing

c  $y = \log_3 x$  approaches  $-\infty$  as  $x$  approaches zero.d  $y$  is undefined.

## Inverses of logarithmic functions

4

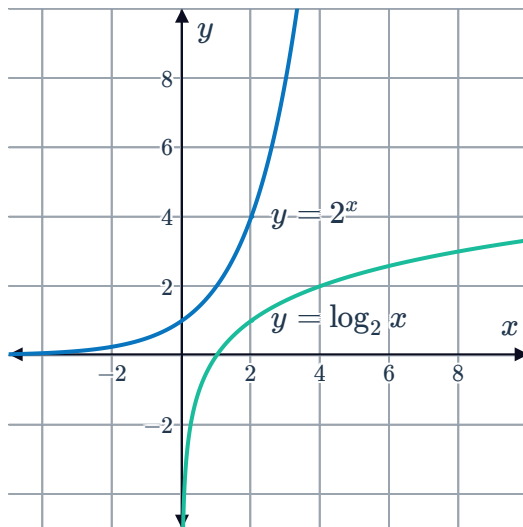
a

|     |   |   |   |   |
|-----|---|---|---|---|
| $x$ | 1 | 2 | 4 | 8 |
| $y$ | 0 | 1 | 2 | 3 |

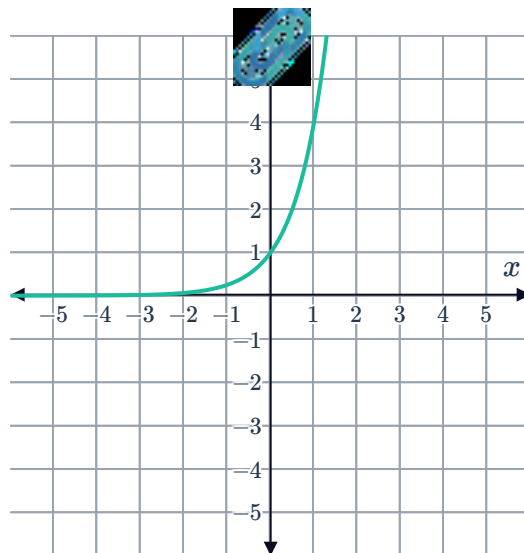
b

|     |   |   |   |   |
|-----|---|---|---|---|
| $x$ | 0 | 1 | 2 | 3 |
| $y$ | 1 | 2 | 4 | 8 |

c

d  $y = 2^x$

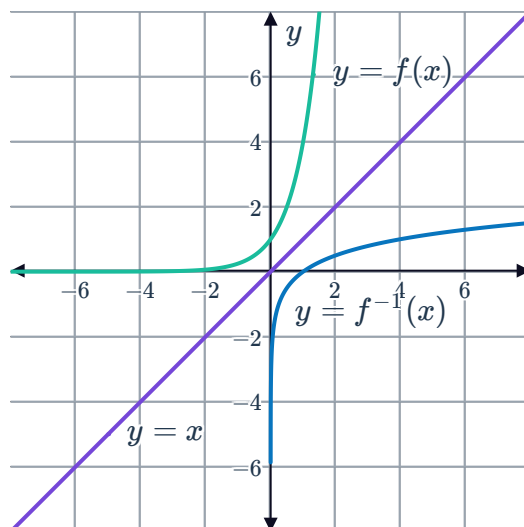
5



6

a

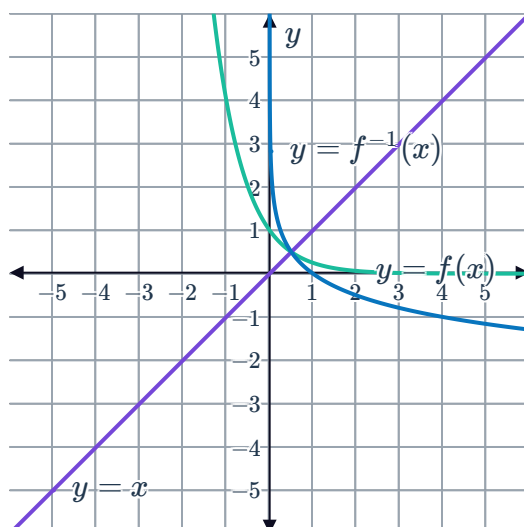
i



ii  $y = \log_4 x$

b

i

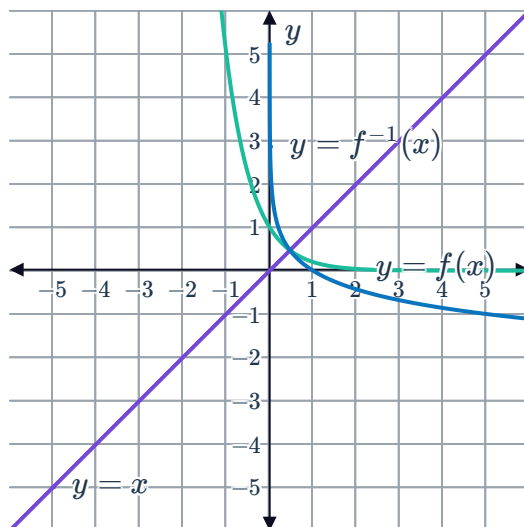


ii  $y = -\log_4 x$

c  
i



ii  $y = \log_{\frac{1}{5}} x$

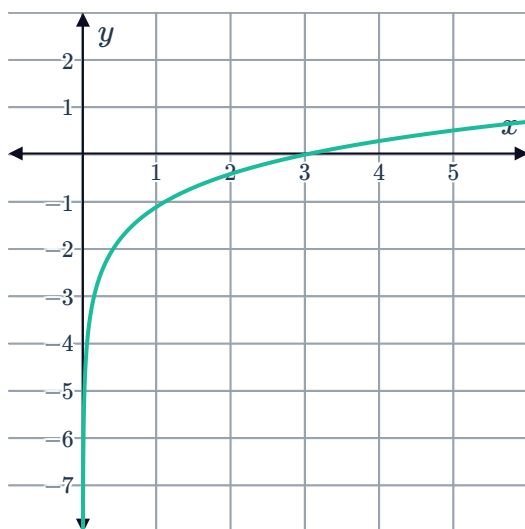


## Natural logarithmic functions

7 The logarithmic function with base  $e$ .

8

a



b  $y = \ln \left( \frac{x}{3} \right)$

9

a  $y = \log_e \left( \frac{x+5}{2} \right)$

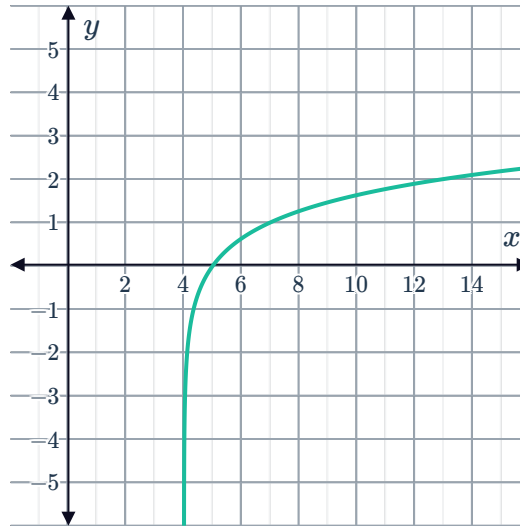
b  $(-4.986, -4.986), (1.118, 1.118)$

## Transformations of logarithmic functions



10  $y = \log_3 (x - 5)$

11

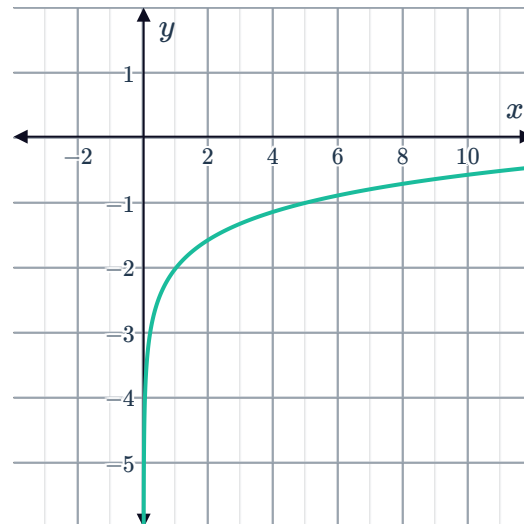


12

a

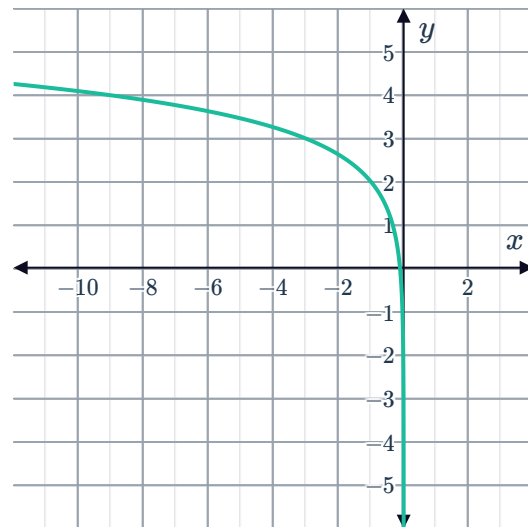
i  $y = \log_5 x - 2$

ii



b

i  $y = \log_3(-x) + 2$



13  $g(x) = 3 \log_5 x$

14

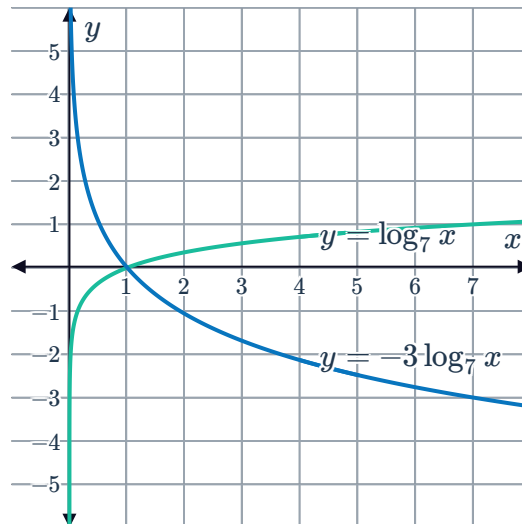
a Reflection over the  $y$ -axis.

b  $g(x) = \ln(-x)$

15

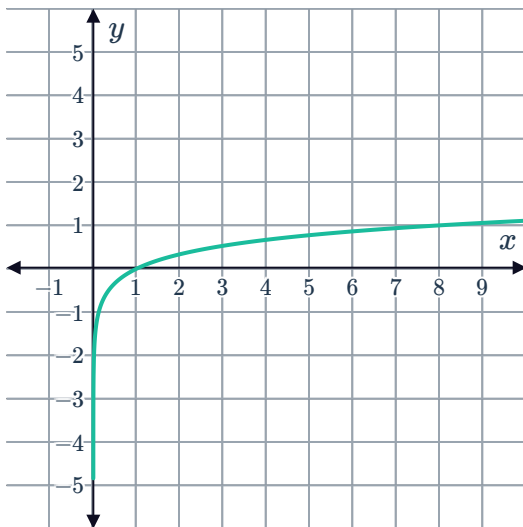
a Vertical stretch by a factor of 3 and reflection across the  $x$ -axis.

b

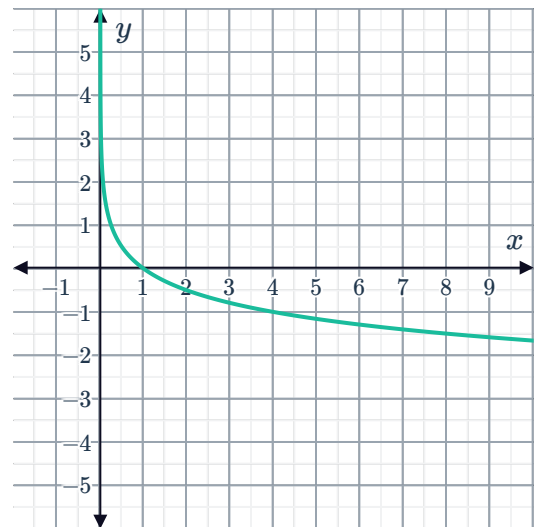


16

a



b



17

a

|              |      |      |                |
|--------------|------|------|----------------|
| $x$          | $-4$ | $-1$ | $-\frac{1}{4}$ |
| $\log_4(-x)$ | $1$  | $0$  | $-1$           |

b

|               |      |     |               |
|---------------|------|-----|---------------|
| $x$           | $-2$ | $1$ | $\frac{7}{4}$ |
| $\log_4(2-x)$ | $1$  | $0$ | $-1$          |

c  $y = \log_4(2-x)$  is a translation of  $y = \log_4(-x)$  by 2 units to the right.

18

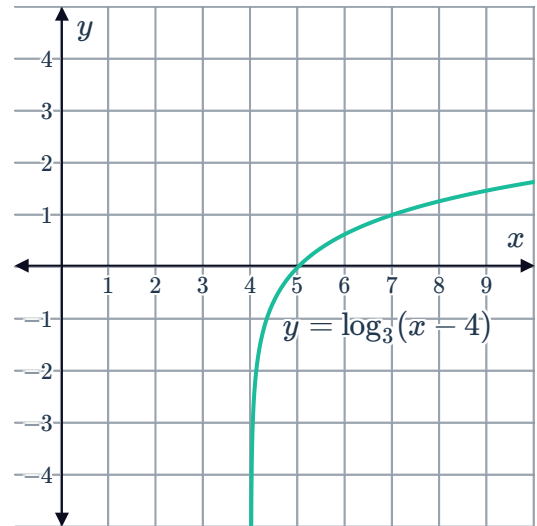
a  $x = 4$

c 1



b  $(5, 0)$

d



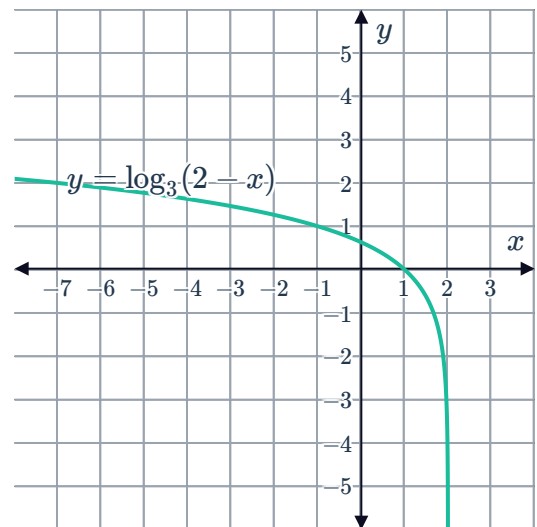
19

a  $x = 2$

c 2

b  $(1, 0)$

d



20

a  $x = 0$

e  $x = -3$

b  $x = 0$

f  $x = 0$

c  $x = 5$

d  $x = 0$



21

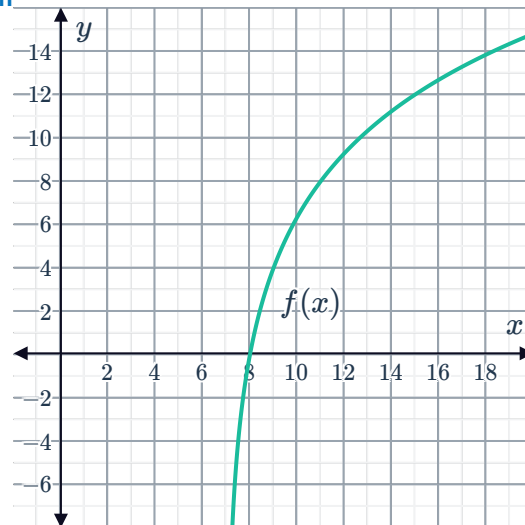
a

i  $x = 8$

ii  $x = 7$



iii

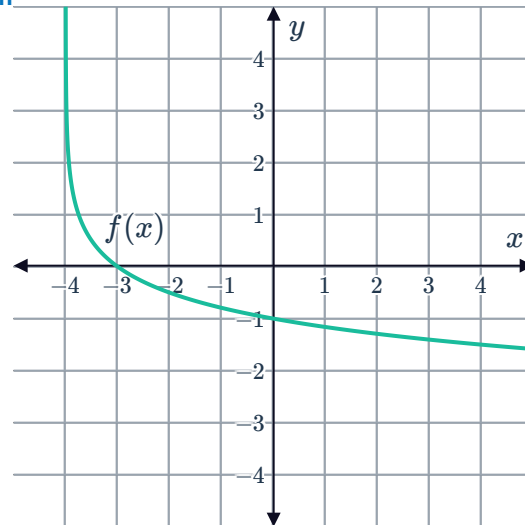


b

i  $x = -3$

ii  $x = -4$

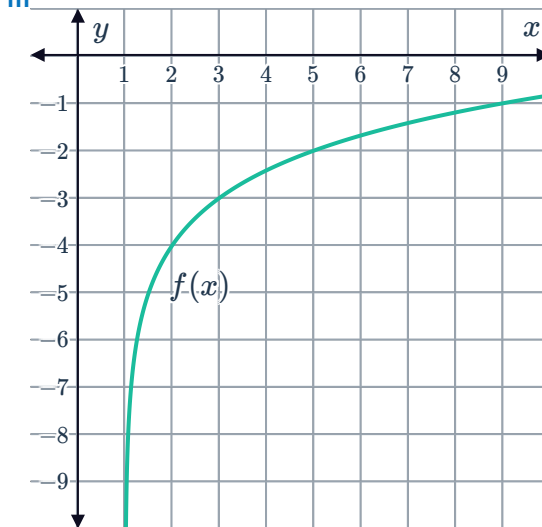
iii



c

i  $x = 17$

ii  $x = 1$

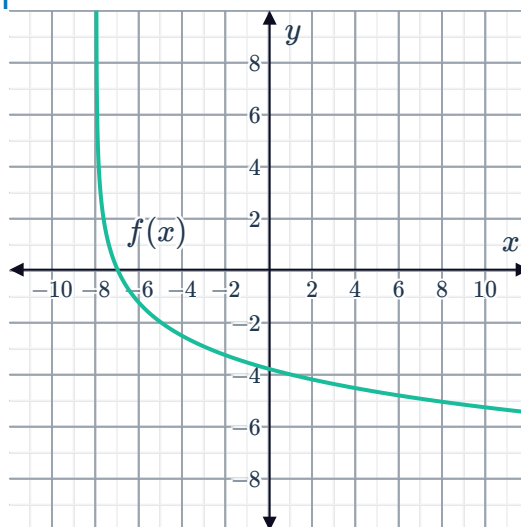


d

i  $x = -7$

ii  $x = -8$

iii

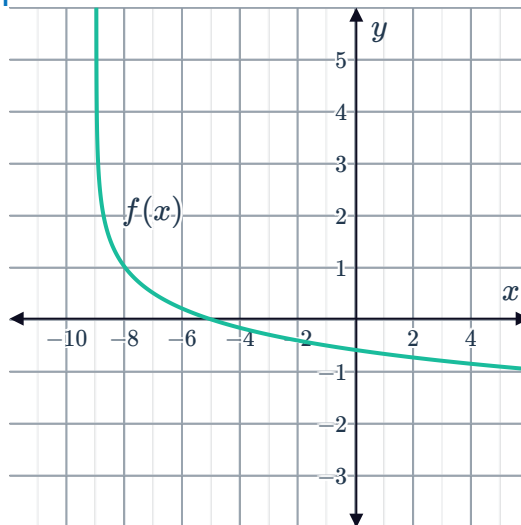


e

i  $x = -5$

ii  $x = -9$

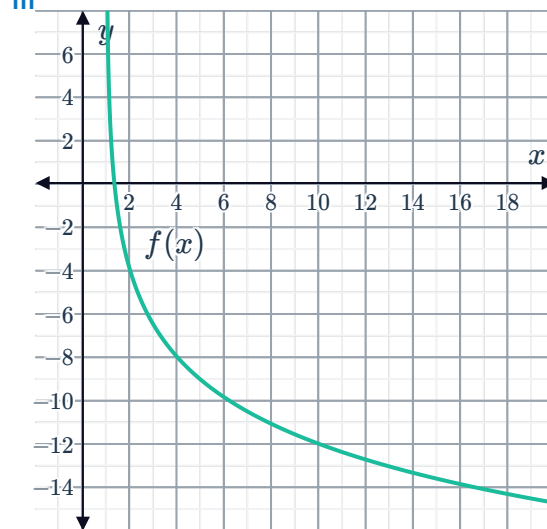
iii



f

i  $x = \frac{4}{3}$

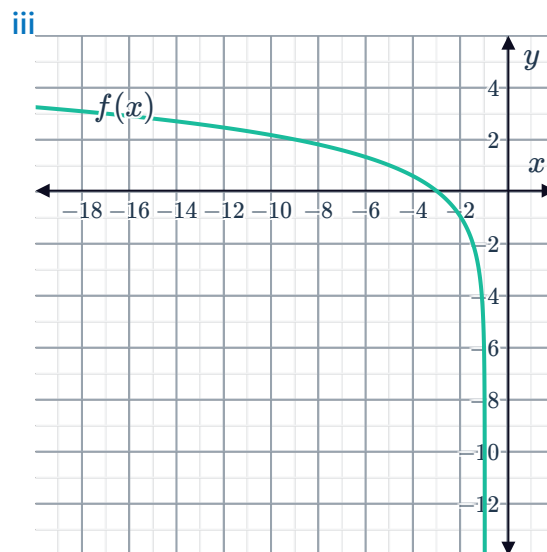
ii  $x = 1$



g

i  $x = -3$

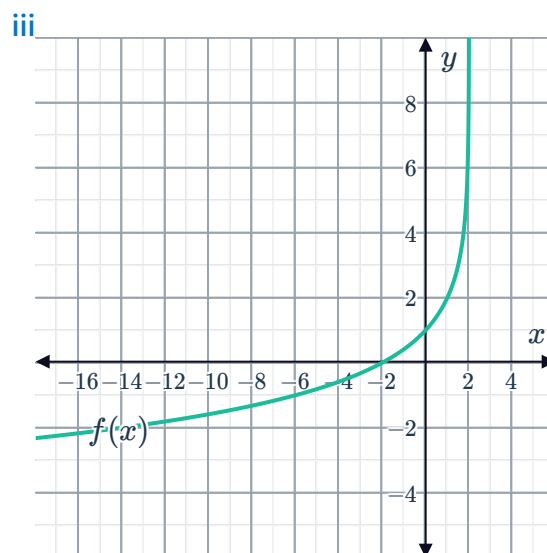
ii  $x = -1$



h

i  $x = -2$

ii  $x = 2$





## Domain and range of logarithmic functions

22

- a  $f(x)$  has been translated 5 units to the right to give  $g(x)$ .  
b The domain changes, but the range remains unchanged.

23

- a  
i  $y = \frac{1}{5}e^x$   
ii  $(-\infty, \infty)$   
iii  $(0, \infty)$
- b  
i  $y = \frac{1}{4}(e^x + 3)$   
ii  $(-\infty, \infty)$   
iii  $\left(\frac{3}{4}, \infty\right)$
- c  
i  $y = \frac{1}{2}\left(e^{\frac{x}{4}} + 5\right)$   
ii  $(-\infty, \infty)$   
iii  $\left(\frac{5}{2}, \infty\right)$
- d  
i  $y = \frac{1}{5}e^{\frac{x+3}{6}}$   
ii  $(-\infty, \infty)$   
iii  $(0, \infty)$
- e  
i  $y = e^{\frac{x+3}{6}} + 4$   
ii  $(-\infty, \infty)$   
iii  $(4, \infty)$
- f  
i  $y = \log_e x - 3$   
ii  $(0, \infty)$   
iii  $(-\infty, \infty)$
- g  
i  $y = \log_e (x + 3)$   
ii  $(-3, \infty)$   
iii  $(-\infty, \infty)$
- h  
i  $y = -\log_e (5 - x)$   
ii  $(-\infty, 5)$   
iii  $(-\infty, \infty)$

i

i  $y = \log_e \left( \frac{x}{2} \right) - 3$



ii  $(0, \infty)$

iii  $(-\infty, \infty)$

j

i  $y = \frac{1}{5} \log_e \left( \frac{x-4}{3} \right)$

ii  $(4, \infty)$

iii  $(-\infty, \infty)$

24

a  $x > 0$

b  $x < 0$

c  $x > 0$

d  $x < 0$

e  $x > -3$

f  $x > -5$

g  $x < 3, x > 7$

h  $-4 < x < 0, x > 4$

i  $x > -5$

j  $x < 3$

k  $-\infty < x < \infty$

l  $-5 < x < 5$